



Subashini M 251801338-2025-AIDS ▾

S2

Started on	Thursday, 17 September 2026, 9:31 AM
State	Finished
Completed on	Thursday, 24 September 2026, 8:20 AM
Time taken	6 days 22 hours
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  #include<stdbool.h>
3
4  bool findPairSum(int arr[], int left, int right, int x, int *element1, int *element2) {
5      if (left >= right) {
6          return false;
7      }
8      int current_sum = arr[left] + arr[right];
9
10     if (current_sum == x) {
11         *element1 = arr[left];
12         *element2 = arr[right];
13         return true;
14     }
15
16     if (current_sum > x) {
17         return findPairSum(arr, left, right - 1, x, element1, element2);
18     } else {
19         return findPairSum(arr, left + 1, right, x, element1, element2);
20     }
21 }
22
23 int main() {
24     int n, x;
25
26     if (scanf("%d", &n) != 1) return 0;
27
28     int arr[n];
29
30     for(int i = 0; i < n; i++) {
31         if (scanf("%d", &arr[i]) != 1) return 0;
32     }
33
34     if (scanf("%d", &x) != 1) return 0;
35
36     int element1 = 0, element2 = 0;
37
38     if (findPairSum(arr, 0, n - 1, x, &element1, &element2)) {
39         printf("%d\n", element1);
40         printf("%d\n", element2);
41     } else {
42         printf("No\n");
43     }
44
45     return 0;
46 }
47

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course